Pedro E. Harunari

Postdoctoral researcher · Statistical physics

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Google Scholar

PERSONAL PROFILE

After obtaining my Ph.D. from the University of São Paulo in 2022 and three years of postdoctoral research in Massimiliano Esposito's group at the University of Luxembourg, I now work with Pierre Ronceray at Aix-Marseille Université on inference from stochastic trajectories.

My research is about inferring hidden dynamics and dissipation from partial, noisy observations of stochastic processes, such as Markov jump processes and Langevin (Itô) dynamics, and about the information-theoretic limits of such inference: Fisher information and Cramér–Rao bounds, Kullback–Leibler divergence estimation, and channel capacity. I look for relations between observable quantities that hold regardless of the hidden details, pin down their thermodynamic limits, and turn them into model-free estimators. Applications range from biophysics and chemical reaction networks to electronics and other complex systems.

EDUCATION AND POSITIONS

Aix Marseille Université — *Marseille, France*
Postdoctoral researcher — *Oct 2025 – Current*

- Supervised by Dr. Pierre Ronceray

University of Luxembourg — *Luxembourg, Luxembourg*
Postdoctoral researcher — *Dec 2022 – Sep 2025*

- Supervised by Prof. Massimiliano Esposito

University of São Paulo — *São Paulo, Brazil*
Doctorate in Physics — *Mar 2018 – Nov 2022*

- "The role of time in nonequilibrium: transition-based coarse-graining, phase transitions and heat engines"
- Advisor: Prof. Dr. Carlos E. Fiore
- Allowed to join the program without a Master's degree
- Approved with the highest grades in every course

University of São Paulo — *São Paulo, Brazil*
Bachelor in Physics — *Feb 2014 – Nov 2017*

- Complementary courses at: IMPA, CBPF and ICTP-SAIFR

University of São Paulo — *São Paulo, Brazil*
Undergraduate researcher — *2015 – 2017*

- Research training program.
- Supervisors: Mário J. de Oliveira (2015–2016), and Carlos E. Fiore (2016–2017).

LIST OF PUBLICATIONS

20 articles in peer-reviewed journals, including Physical Review Letters (Editors' Suggestion), Physical Review X, SciPost Physics, Letters in Mathematical Physics and Journal of Statistical Mechanics, plus 3 preprints.
Google Scholar: 568 citations, h-index 13.

[1] **PE Harunari**, G Fiusa, M Polettini, "*Mutual linearity of complexes in chemical reaction networks*", arXiv:2610.11970 (preprint) **2026**

[2] N Freitas, **PE Harunari**, M Esposito, "*Thermodynamic efficiency of communication channels*", arXiv:2609.25934 (preprint) **2026**

[3] G Fiusa, **PE Harunari**, AJB Rosal, JM Nichol, GT Landi, "*Stochastic trajectories and excursions in a double quantum dot system*", arXiv:2605.20166 (preprint) **2026**

[4] G Fiusa, **PE Harunari**, AS Hegde, GT Landi, "*A framework for fluctuating times and counting observables in stochastic excursions*", Physical Review E **114**, 024129 **2026**

[5] G Fiusa, **PE Harunari**, AS Hegde, GT Landi, "*Counting observables in stochastic excursions*", Physical Review E **114**, L022102 **2026**

[6] S Dal Cengio, **PE Harunari**, V Lecomte, M Polettini, "*Mutual Multilinearity of Nonequilibrium Network Currents*", SciPost Physics **19**, 111 **2025**

[7] M Polettini, **PE Harunari**, S Dal Cengio, V Lecomte, "*Coplanarity of rooted spanning-tree vectors*", Letters in Mathematical Physics **116**, 1 **2025**

[8] Q Yu, **PE Harunari**, "*Dissipation at limited resolutions: Power law and detection of hidden dissipative scales*", Journal of Statistical Mechanics: Theory and Experiment **2024**, 103201 **2024**

[9] **PE Harunari**, CE Fiore, AC Barato, "*Inference of entropy production for periodically driven systems*", Physical Review E **110**, 064126 **2024**

★ [10] **PE Harunari**, S Dal Cengio, V Lecomte, M Polettini, "*Mutual linearity of nonequilibrium network currents*", Physical Review Letters **133**, 047401 (Editors' Suggestion) **2024**

[11] **PE Harunari**, "*Uncovering Nonequilibrium from Unresolved Events*", Physical Review E **110**, 024122 **2024**

[12] A Garilli, **PE Harunari**, M Polettini, "*Fluctuation relations for a few observable currents at their own beat*", Journal of Physics A: Mathematical and Theoretical **57**, 455003 **2024**

[13] F Avanzini, M Bilancioni, V Cavina, S Dal Cengio, M Esposito, G Falasco, D Forastiere, N Freitas, A Garilli, **PE Harunari**, V Lecomte, A Lazarescu, SGM Srinivas, C Moslonka, I Neri, E Penocchio, WD Piñeros, M Polettini, A Raghu, P Raux, K Sekimoto, A Soret, "*Methods and Conversations in (Post)Modern Thermodynamics*", SciPost Physics Lecture Notes **80** **2024**

[14] F Hawthorne, **PE Harunari**, MJ de Oliveira, CE Fiore, "*Nonequilibrium thermodynamics of the majority vote model*", Entropy **25**, 1230 (Feature Paper) **2023**

[15] **PE Harunari**, A Garilli, M Polettini, "*Beat of a current*", Physical Review E **107**, L042105 **2023**

★ [16] **PE Harunari**, A Dutta, M Polettini, É Roldán, "*What to learn from a few visible transitions' statistics?*", Physical Review X **12**, 041026 **2022**

[17] IN Mamede, **PE Harunari**, BAN Akasaki, K Proesmans, CE Fiore, "*Obtaining efficient thermal engines from interacting Brownian particles under time-periodic drivings*", Physical Review E **105**, 024106 **2022**

[18] CE Fiore, **PE Harunari**, CEF Noa, GT Landi, "*Current fluctuations in nonequilibrium discontinuous phase transitions*", Physical Review E **104**, 064123 **2021**

[19] **PE Harunari**, S Fernando Filho, CE Fiore, A Rosas, "*Maximal power for heat engines: Role of asymmetric interaction times*", Physical Review Research **3**, 023194 **2021**

[20] **PE Harunari**, CE Fiore, K Proesmans, "*Exact statistics and thermodynamic uncertainty relations for a periodically driven electron pump*", Journal of Physics A: Mathematical and Theoretical **53**, 374001 **2020**

[21] CEF Noa, **PE Harunari**, MJ de Oliveira, CE Fiore, "*Entropy production as a tool for characterizing nonequilibrium phase transitions*", Physical Review E **100**, 012104 **2019**

[22] JM Encinas, **PE Harunari**, MM de Oliveira, CE Fiore, "*Fundamental ingredients for discontinuous phase transitions in the inertial majority vote model*", Scientific Reports **8**, 9338 **2018**

[23] **PE Harunari**, MM de Oliveira, CE Fiore, "*Partial inertia induces additional phase transition in the majority vote model*", Physical Review E **96**, 042305 **2017**

RESEARCH VISITS AND WORKING GROUPS

Workshop: Stochastic Thermodynamics and Computer Science Theory, UT Austin *Austin, USA*
Invited participant, invitation-only working group *2026*

Santa Fe Institute *Santa Fe, USA*
Invited participant, working group NeST: Neuromorphic Stochastic Thermodynamics *2025*

Santa Fe Institute *Santa Fe, USA*
Invited participant, working group: Stochastic Thermodynamics and Computer Science Theory *Jun 2024*

The Abdus Salam International Centre for Theoretical Physics (ICTP) *Trieste, Italy*
Visiting researcher *May 2022 – Jun 2022*

- Visitor at Édgar Roldán's group.

University of Luxembourg *Luxembourg, Luxembourg*
Visiting researcher *Apr 2021 – Feb 2022*

- Visitor at Massimiliano Esposito's Complex Systems and Statistical Mechanics group.
- Supervised by Matteo Polettini.

University of Aalto *Helsinki, Finland*
Visiting researcher *Dec 2021 – Jan 2022*

- Visitor at Jukka Pekola's PICO group.

The Abdus Salam International Centre for Theoretical Physics (ICTP) *Trieste, Italy*
Visiting researcher *Jul 2021 – Sep 2021*

- Visitor at Édgar Roldán's group.

EVENT ORGANIZATION

(Post)Modern Thermodynamics - School and workshop *Luxembourg, Luxembourg*
Organizer *Dec 2022*

- Approximately, 100 participants from abroad and 30 from Luxembourg
- Conference consisting of 10 school lectures, 8 workshop sessions, and one poster session.
- Shared teaching duties of the lecture "*Continuous-time Markov chain: basics, first-passages and thermodynamics*" with Ken Sekimoto.
- Editorial duties on the preparation of lecture notes "Methods and Conversations in (Post)Modern Thermodynamics".
- Co-organizers: Matteo Polettini, Vasco Cavina, William Piñeros.

Statistical Physics seminar series *online*
Organizer *2020*

- 21 seminars virtually presented during the COVID lockdown, mostly by professors, for a broad audience of students and researchers across Brazil and other countries. Co-organizer: Carlos E. Fiore.

TEACHING EXPERIENCE

2024 **Classical and Quantum Information Theory (graduate level)**, lecturer, University of Luxembourg
2020 **Thermodynamics**, teaching assistant, University of São Paulo
2019 **Statistical Mechanics**, teaching assistant, University of São Paulo
2018 **Statistical Mechanics (graduate level)**, teaching assistant, University of São Paulo
2018 **Statistical Mechanics**, teaching assistant, University of São Paulo

SKILLS

Theory Stochastic processes and stochastic calculus (Markov jump processes and master equations, Langevin/Itô SDEs, first-passage and counting statistics), large deviations, nonequilibrium statistical mechanics.
Inference Maximum likelihood, Fisher information and Cramér–Rao bounds, model-free estimators from partial or noisy data, information theory (KL divergence, mutual information, channel capacity).
Numerics Stochastic simulation (Gillespie, SDE integration), gradient-based optimization with automatic differentiation, spectral and graph-theoretic methods for Markov generators.
Programming Python (JAX, NumPy, SciPy, Matplotlib), Mathematica, C; git, HPC clusters.

PRIZES AND GRANTS

2023 **Honorable mention for distinguished Ph.D. thesis in Exact and Earth Sciences**, University of São Paulo

2023 **Best Ph.D. thesis in Statistical and Computational Physics (national level)**, Brazilian Physical Society

2022 **Best Ph.D. thesis prize**, Institute of Physics - University of São Paulo

2022 **Honorable mention for distinguished publication**, Institute of Physics - University of São Paulo

2021 **Internship grant**, grant for 11 months of internship abroad (BEPE - FAPESP)

2018 **Ph.D. fellowship**, FAPESP 4 years grant for the Doctorate without Master's degree program

2017 **Undergrad research fellowship**, FAPESP grant for the Undergraduate research program

2016 **Undergrad research fellowship**, CNPq grant for the Undergraduate research program

INVITED TALKS AND SEMINARS

2027 IntCha - Interdisciplinary Challenges in the Physics of Complexity and Life, University of Padova, Padova, Italy (invited conference talk, to occur)

2026 "*Mutual linearity: rigid response far from equilibrium*", Complex Systems Seminar, University of Rochester, USA

2026 "*The capacity of learning through a noisy channel*", Workshop: Stochastic Thermodynamics and Computer Science Theory, UT Austin, Austin, USA (invited talk)

2026 University of Strasbourg, France (to occur)

2026 "*Recent unfoldings on Mutual Linearity*", Univ. of São Paulo, Brazil

2026 "*Stochastic excursions: from counting observables to inference*", ICTP, Italy

2026 "*Stochastic excursions: Counting observables and noise*", Univ. of São Paulo, Brazil

2026 "*Stochastic Excursions: Counting observables and noise*", SigmaPhi International Conference on Statistical Physics, Workshop "Statistical Physics of Complex Systems", Kolymbari, Greece (invited conference talk)

2025 "*Thermodynamic cost of communication channels*", Universitat Pompeu Fabra, Spain

2025 "*Thermodynamic limits of nonequilibrium communication across scales*", Princeton University, USA

2025 "*Thermodynamic limits of nonequilibrium communication across scales*", Massachusetts Institute of Technology (MIT)

2025 "*Stochastic Thermodynamics of communication channels*", Santa Fe Institute, USA

2024 "***Colloquium:*** *Recuperando termodinâmica estocástica com informação limitada*", Univ. of São Paulo, Brazil

2024 "*Mutual Linearity of Nonequilibrium Network Currents*", Université Paris Cité, France

2024 "*Mutual Linearity of Nonequilibrium Network Currents*", Aix-Marseille Université, France

2024 "*Model-free inference of entropy production from partial information*", Université Grenoble Alpes, France

2021 "*Inferences from partial information of transition statistics*", University of Aalto, Finland

2019 "*Stochastic Thermodynamics: basics and some modern aspects*", *"Coloquinho"*, series of talks organized by IFT-UNESP, ICTP-SAIFR students, São Paulo, Brazil

2019 "*Einstein's contributions to Statistical Mechanics*", Physics Giants: Einstein Week, series of talks organized by USP students, São Paulo, Brazil

CONTRIBUTED TALKS AND POSTERS

2026 **Poster**: "*Thermodynamic Limits of Communication Channels*", Japan-France joint seminar : Statistical Physics Out of Equilibrium, Institut Henri Poincaré, Paris, France

2026 **Poster**: "*Mutual Linearity of Currents*", IntCha - Interdisciplinary Challenges in the Physics of Complexity and Life, Institut d'Études Scientifiques de Cargèse, Cargèse, France

2025 **Talk**: "*Thermodynamic limits of communication channels*", Workshop on Non-equilibrium Statistical Physics, ICTP - SAIFR, São Paulo, Brazil

2025 **Poster**: "*Thermodynamic limits of nonequilibrium communication across scales*", STATPHYS 29, The 29th international conference on statistical physics, Florence, Italy

2025 **Poster**: "*Mutual linearity of currents*", Stochastic processes: Inferences in complex systems, CECAM, Lausanne, Switzerland

2025 **Talk**: "*Mutual linearity of nonequilibrium network currents*", Workshop on Stochastic Thermodynamics - WOST VI, virtual conference, online

2024 **Poster**: "*Inferring dissipation by monitoring reservoirs*", Dissipative Processes in Molecular Systems, University of Padova, Padova, Italy

2024 **Talk**: "*Unveiling nonequilibrium from multifilar events*", Journées de Physique Statistique, 42nd edition, École Normale Supérieure de Paris, Paris, France

2023 **Talk**: "*Thermodynamics at the beat of transitions*", XXVII Sitges Conference on Statistical Mechanics, Universitat de Barcelona, Sitges, Spain

2023 **Talk**: "*Fluctuation relation at the beat of a current*", Workshop on Stochastic Thermodynamics - WOST IV, The Abdus Salam International Centre for Theoretical Physics (ICTP), online

2023 **Talk**: "*Thermodynamics through the lens of transitions*", Physics of Life: Students and Postdocs Edition, The Center for the Physics of Biological Function, CUNY/Princeton, New York, United States of America

2023 **Talk**: "*Thermodynamics at the beat of transitions*", Fluctuations and First Passage Problems, NORDITA, Stockholm, Sweden

2023 **Talk**: "*Transition-based coarse-graining*", Journées de Physique Statistique, 42nd edition, École Normale Supérieure de Paris, Paris, France

2022 **Poster**: "*Inferences from Statistics of a Few Observable Transitions*", The 47th Conference of the Middle European Cooperation in Statistical Physics, MECO 47, Erice, Italy

2022 **Poster**: "*Inferences from Statistics of a Few Observable Transitions*", Autumn meeting, Brazilian Physical Society, São Paulo, Brazil

2021 **Talk**: "*Inferences from Statistics of a Few Observable Transitions*", National Statistical Physics Meeting, Universidade Federal de São João del-Rei (UFSJ), online

2021 **Poster**: "*Entropy production fluctuation in phase transitions*", Statistical Physics of Complex Systems, The Abdus Salam International Centre for Theoretical Physics (ICTP), Trieste, Italy

2021 **Talk**: "*Entropy Production fluctuations in nonequilibrium transitions*", Autumn meeting, Brazilian Physical Society, online

2021 **Talk**: "*Quantitative comparison of different time-periodic Thermodynamic Uncertainty Relations*", APS March Meeting, American Physical Society, online

2020 **Talk**: "*Stochastic Thermodynamics: Schnakenberg, FT and TUR*", Statistical Physics Seminar Series, University of São Paulo, online

2020 **Talk**: "*Stochastic pump as a model to study nonequilibrium properties*", Autumn meeting, Brazilian Physical Society, online

2020 **Poster**: "*Time asymmetric reciprocity relations for an arbitrarily long single-particle stochastic pump and its exact solution*", Quantum Thermodynamics for Young Scientists, Wilhelm and Else Heraeus-Foundation, Bad Honnef, Germany

Also attended: Bangalore School on Statistical Physics XII (International Center for Theoretical Sciences (ICTS), online, 2021); Workshop on Stochastic Thermodynamics - WOST II (Santa Fe Institute, online, 2021); Quantum Thermodynamics of Non-equilibrium systems (Donostia International Physics Center, online, 2020).

LANGUAGES

English	Full professional proficiency
Portuguese	Native proficiency
French	Conversational
Spanish	Limited working proficiency

PEER-REVIEW CONTRIBUTIONS

- Physical Review Letters
- Physical Review X
- Physical Review X Quantum
- Physical Review E
- Physical Review A
- New Journal of Physics
- Journal of Statistical Mechanics: Theory and Experiment
- npj Unconventional Computing
- Journal of Physics A: Mathematical and Theoretical
- Journal of Physics Communications

REFERENCES

Available upon request.